

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

### ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

|                         |                                                                                                                    |                      |                                             |
|-------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------|
| <b>Course Code:</b>     | CSA0309                                                                                                            | <b>Course Title:</b> | Data Structures                             |
| <b>CO Assessed:</b>     | CO5 – Naturalization (BL4, BL6)                                                                                    | <b>Assessment:</b>   | Assessment Tool 1 – Pseudocode Writing Task |
| <b>Weightage:</b>       | 35% of CIA                                                                                                         | <b>Total Marks:</b>  | 100                                         |
| <b>CO5 Statement:</b>   | <i>Construct MST using shortest path algorithms and traverse the graph to model and analyse realworld problem.</i> |                      |                                             |
| <b>Student Name:</b>    | Rachepalli Rohini Sree                                                                                             | <b>Reg. No.:</b>     | 192524471                                   |
| <b>Section / Batch:</b> | 2 <sup>nd</sup> year                                                                                               | <b>Date:</b>         | 28/08/2026                                  |

#### Code of Conduct

I \_\_\_Rachepalii Rohini sree\_(192524471)\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: \_\_\_Rohini sree\_

#### Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

| Section / Topic         | Focus                      | Q Nos | Marks |
|-------------------------|----------------------------|-------|-------|
| Minimum Spanning Tree   | Kruskal's Algorithm        | 1     | 20    |
| Graph Sorting           | Topological Sort           | 2     | 20    |
| Shortest Path Algorithm | MST & Dijkstra's Algorithm | 3     | 20    |

|                                |                               |                  |    |
|--------------------------------|-------------------------------|------------------|----|
| <b>Shortest Path Algorithm</b> | Dijkstra's Algorithm          | 4                | 20 |
| <b>Minimum Spanning Tree</b>   | Prim's or Kruskal's Algorithm | 5                | 20 |
| <b>GRAND TOTAL:</b>            |                               | <b>100 Marks</b> |    |

### Rubrics for Calculation

| Criteria                              | Weightage | Level 4 – Exemplary                                                           | Level 3 – Proficient                                 | Level 2 – Developing                            | Level 1 – Beginning               |
|---------------------------------------|-----------|-------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------|-----------------------------------|
| <b>Problem Understanding</b>          | 20        | Demonstrates complete and clear understanding; handles edge cases effectively | Good understanding; minor gaps in edge case handling | Partial understanding; misses some requirements | Misinterprets problem; major gaps |
| <b>Code Correctness</b>               | 30        | Fully correct; passes all test cases                                          | Mostly correct; minor errors                         | Partially correct; several test cases fail      | Incorrect or nonfunctional code   |
| <b>Code Quality &amp; Readability</b> | 20        | Clean, modular, wellcommented, follows standards                              | Readable with some structure                         | Limited readability; poor structure             | Unorganized and hard to read      |
| <b>Complexity Analysis</b>            | 20        | Accurate time & space complexity with justification                           | Correct but limited explanation                      | Incomplete or partially correct                 | Missing or incorrect              |
| <b>Testing &amp; Validation</b>       | 10        | Thorough testing including edge cases                                         | Adequate testing with few edge cases                 | Minimal testing                                 | No testing evidence               |
| <b>Faculty In-charge</b>              |           | <b>Course Coordinator</b>                                                     |                                                      | <b>Program Director</b>                         |                                   |
| Signature: _____                      |           | Signature: _____                                                              |                                                      | Signature: _____                                |                                   |
| Date: _____                           |           | Date: _____                                                                   |                                                      | Date: _____                                     |                                   |

## QUESTIONS

- Q1** Minimum Spanning Tree - Kruskal's Algorithm  
20 marks
- Q2** Graph - Topological Sort  
20 marks
- Q3** Shortest Path Algorithm - MST & Dijkstra's Algorithm  
20 marks
- Q4** Shortest Path Algorithm - Dijkstra's Algorithm  
20 marks
- Q5** Minimum Spanning Tree - Prim's or Kruskal's Algorithm  
20 marks

5/5 answered

**Q1** Minimum Spanning Tree - Kruskal's Algorithm

20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

- a. All zones must be connected
- b. Total installation cost must be minimized
- c. No redundant connections (no cycles) allowed
- d. The network must remain fully connected

## Your Answer

START

Input the graph with vertices and weighted edges.  
sort all edges in increasing order of their weights.

Initialize an empty set MST.

For each edge(u, v) in sorted order:

check whether adding(u, v) creates a cycle.

Save Answer

Next →

## QUESTIONS

- Q1** Minimum Spanning Tree - Kruskal's Algorithm  
20 marks
- Q2** Graph - Topological Sort  
20 marks
- Q3** Shortest Path Algorithm - MST & Dijkstra's Algorithm  
20 marks
- Q4** Shortest Path Algorithm - Dijkstra's Algorithm  
20 marks
- Q5** Minimum Spanning Tree - Prim's or Kruskal's Algorithm  
20 marks

5/5 answered

**Q2** Graph - Topological Sort

20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.

Constraints:

- a. Detect cycle before sorting
- b. Use DFS approach

## Your Answer

START

INPUT the graph with vertices and weighted edges.  
Sort all edges in increasing order of their weights.

Initialize an empty set "MST".

For each edge "(u, v)" in sorted order:

- check whether adding "(u, v)" creates a cycle.

Save Answer

Next →

## QUESTIONS

Q1  
Minimum Spanning Tree -  
Kruskal's Algorithm  
20 marks

Q2  
Graph - Topological Sort  
20 marks

Q3  
Shortest Path Algorithm - MST  
& Dijkstra's Algorithm  
20 marks

Q4  
Shortest Path Algorithm -  
Dijkstra's Algorithm  
20 marks

Q5  
Minimum Spanning Tree -  
Prim's or Kruskal's Algorithm  
20 marks

5/5 answered

## Q3 Shortest Path Algorithm - MST &amp; Dijkstra's Algorithm

20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.

Constraints:

- Use MST for infrastructure
- Use shortest path for routing

## Your Answer

Algorithm Smart Transportation(G, Source ,Destination)  
sort all roads by cost.  
Apply Kruskal's algorithm to construct MST.  
Set distance[source]=0 and all others =infinity.  
Apply Dijkstra's algorithm on the MST.  
Select the shortest path from Source to Destination.

Save Answer

Next →

## QUESTIONS

Q1  
Minimum Spanning Tree -  
Kruskal's Algorithm  
20 marks

Q2  
Graph - Topological Sort  
20 marks

Q3  
Shortest Path Algorithm - MST  
& Dijkstra's Algorithm  
20 marks

Q4  
Shortest Path Algorithm -  
Dijkstra's Algorithm  
20 marks

Q5  
Minimum Spanning Tree -  
Prim's or Kruskal's Algorithm  
20 marks

5/5 answered

## Q4 Shortest Path Algorithm - Dijkstra's Algorithm

20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

## Your Answer

Algorithm Toll FreeShortestpath(G,S,D)  
Set all other distances = infinity.  
Apply Dijkstra's algorithm.  
For each edge(u, v, w):  
a. Move normally with cost w.  
b. IF voucher is unused, move with cost 0 and mark voucher as used.

Save Answer

Next →

## QUESTIONS

Q1  
Minimum Spanning Tree -  
Kruskal's Algorithm  
20 marks

Q2  
Graph - Topological Sort  
20 marks

Q3  
Shortest Path Algorithm - MST  
& Dijkstra's Algorithm  
20 marks

Q4  
Shortest Path Algorithm -  
Dijkstra's Algorithm  
20 marks

Q5  
Minimum Spanning Tree -  
Prim's or Kruskal's Algorithm  
20 marks

5/5 answered

## Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm

20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST. (20 Marks)

Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

## Your Answer

Algorithm Second Best MST(G)

Find the MST of G using Kruskal's algorithm.

Set Second Best = infinity.

For each edge  $e$  in MST:

Remove  $e$  from MST.

Find the minimum -cost edge that reconnects the two components.

[Save Answer](#)